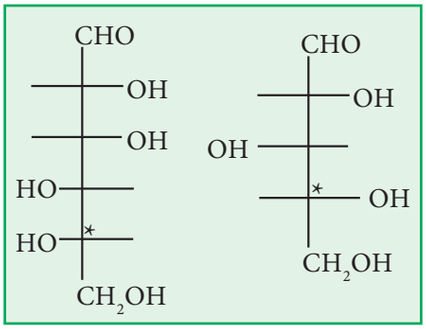
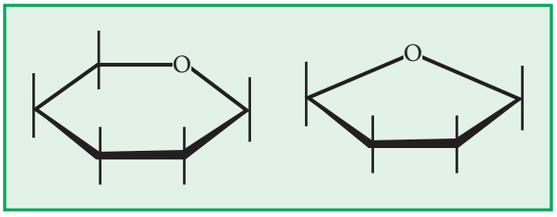
5.5 Haworth's Projection formula

A Haworth Projection is a common way of writing a structural formula to represent the cyclic structure of monosaccharides. It was named in the remembrance of chemist Sir Norman Haworth. Let's consider two sugars, one an aldohexose, the other an aldopentose. Fischer projections are shown below. Recall the carbon with the star (*) next to it is the one that determines if the sugar is a D -sugar or an L-sugar.

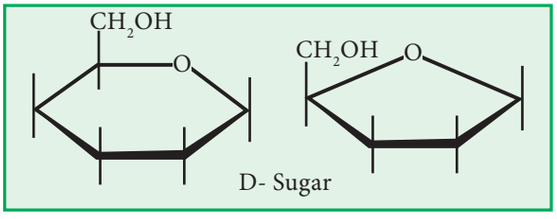


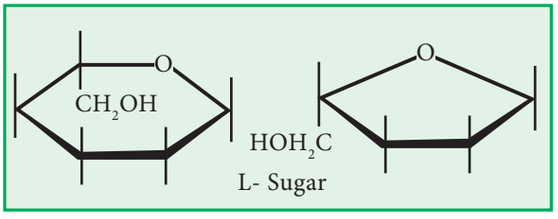
The Following are the rules.

1. Draw the basic structure for the sugar.

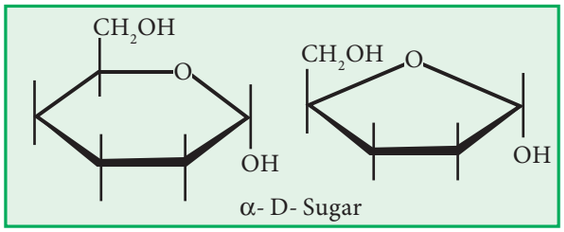


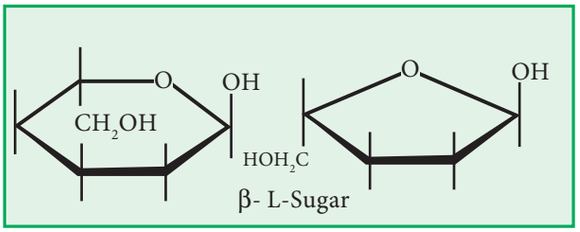
2. If the sugar is a D-sugar, place a $-CH_2OH$ above the ring on the carbon to the left of the oxygen, for an L-sugar place it below the ring.





3. For an -sugar place an α- OH below the ring on the carbon to the right of the ring oxygen, for an β-sugar place the - OH above the ring.





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